Pointer Arithmetic and Arrays

Pointer arithmetic involves performing operations on pointers, such as addition, subtraction, and comparison. It's closely related to arrays because arrays and pointers share a deep connection in C.

Basics of Pointer Arithmetic

- **Incrementing a Pointer:** Moves the pointer to the next memory location based on its data type size.

```
int arr[3] = {10, 20, 30};

int *ptr = arr;

ptr++;  // Points to the next element (20)
```

- **Decrementing a Pointer:** Moves the pointer to the previous memory location.
- **Pointer Addition/Subtraction:** Adding or subtracting an integer from a pointer shifts it by that many elements.

Pointers and Arrays Relationship

Arrays are essentially pointers to the first element. Therefore, `arr` and `&arr[0]` are equivalent. Example:

```
int arr[3] = {1, 2, 3};

int *ptr = arr;
```